

Goal

Object detection with a bounding box on all objects. For every object, pretrained and prelabelled datasets.

Objective

To develop a comprehensive object detection system that surpasses traditional COCO-based models by leveraging the extensive and diverse **Open Images V7** dataset (600+ object categories).

The system will be designed to detect **real-world objects in real-time**, using **Grounding DINO + Segment Anything (SAM)** models and webcam input.

Background

Traditional object detection systems like YOLOv5 and YOLOv8 are limited to 80 classes (e.g., COCO). However, Open Images V7 provides a richer dataset with:

- **9 million images**
- **600+ boxable object categories**
- **Real-world annotations** including bounding boxes, segmentation masks, and relationships.

We aim to utilize this richness to build a universal detection app.

Methodology

1. Dataset Preparation

- Use Google's official **Open Images V7 dataset**.
- Download metadata CSVs (classes, annotations, image URLs).
- Filter and download relevant subset (e.g., top 100 classes).
- Convert annotation format to YOLOv5-compatible format.

2. Modeling Approach

- Use pre trained **Grounding DINO** to handle open-vocabulary detection.
- Integrate **Segment Anything Model (SAM)** for visual segmentation.
- Optionally fine-tune YOLOv5/YOLOv8 on Open Images subset.

3. Live Detection System

- Stream real-time video from webcam.
- Deploy object detection + segmentation pipeline.
- Use **Streamlit** or a Python GUI for interface.

Deliverables

- Fully working app detecting 100+ real-world object types via webcam.
- Detection accuracy metrics.
- Comparison with COCO-trained models (YOLOv5 baseline).
- Presentation slides + live demo.
- Google Colab notebook and GitHub repository.

Timeline

Week	Milestone
1	Dataset filtering and annotation formatting
2	Grounding DINO + SAM model integration
3	Streamlit-based live webcam testing
4	Evaluation, visualization, and final report

Innovation

This project merges **open-vocabulary detection** (Grounding DINO) with **universal segmentation** (SAM) using a rich public dataset (Open Images V7),

moving beyond rigid label-based object detection. The approach is scalable to any object type—real-world tools, logos, scenes, etc.